

Bentley®
Advancing Infrastructure

 **CONNECT Edition**

Key Components

Modeling

- Precede Graphical Modeler
- Datagen Intelligent Editor
- Gap Elements
- Superelements

Loads

- Seastate Wave, Wind, Current
- Buoyancy, Mud flow
- Gravity, Inertial
- Skid, Moving

Analysis

- SACS IV Solver
- Large Deflection (LDF)
- PSI Pile/Structure Analysis
- Liquefaction
- Wind Turbine Analysis
- Dynpac Modal Analysis
- Collapse
- Collapse Advanced

Design

- Combine Solution Files
- Post Offshore Code Design
- Concrete Design
- Postvue Graphical Redesign
- Joint Can
- Interactive Fatigue
- Dynamic Fatigue
- Wave Fatigue



SACS™

Design and Analysis Software for Offshore Structures

SACS is an integrated finite element structural analysis package of applications that uniquely provides for the design of offshore structures, including oil and gas platforms, wind farms, and topsides of FPSOs and floating platforms. SACS software has been used by offshore engineers around the world for nearly 40 years. Many of the world's energy companies specify SACS software for use by their engineering firms across the lifecycle of offshore platforms. Because every project has different needs, there are three flexible SACS options. Offshore Structure Ultimate for comprehensive capabilities required for typical offshore jackets, wharfs, and dolphin structures; Offshore Structure Advanced for static topside and deck analysis; and Offshore Structure for static structural analysis.

Offshore Structure Ultimate

Offshore Structure Ultimate contains capabilities required for typical offshore jackets, wharfs, and dolphin structures. It includes the interactive graphics modeler with advanced 3D capabilities, SACS IV solver and interactive graphics 3D post processor, Seastate, Joint Can, Pile, Combine, Gap, Tow, and LDF large deflection. The package also includes automatic model generation, beam and finite element capability, steel code check and redesign, environmental load generation, tubular connection check, single pile/soil interaction, inertia and moving load generation, tension/compression nonlinear elements with initial gap, load case combination, linear large deflection analysis, and full output report and plotting capabilities. The package also contains the multicore analysis capability allowing the user to conduct multiple analyses of the same type in parallel, saving hours of runtime.

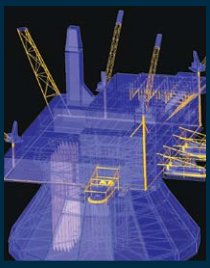
Offshore Structure Advanced

Offshore Structure Advanced contains capabilities required for typical topside and deck analysis. It includes the interactive graphics modeler with advanced 3D capabilities, SACS IV solver

and interactive graphics 3D post-processor, topsides wind loading, Combine, Gap, Tow, and LDF large deflection. The package also includes automatic model generation, beam and finite element capability, steel code check and redesign, wind and gravity load generation, inertia and moving load generation, tension/compression nonlinear elements with initial gap, load case combination, linear large deflection analysis, and full output report and plotting capabilities.

Offshore Structure

Offshore Structure contains capabilities required for static structural analysis. It includes the interactive graphics modeler with advanced 3D capabilities, SACS IV solver and interactive graphics 3D post-processor, Combine, Gap, Tow, and LDF large deflection. The package also includes automatic model generation, beam and finite element capability, steel code check and redesign, inertia and moving load generation, tension/compression nonlinear elements with initial gap, load case combination, linear large deflection analysis, and full output report and plotting capabilities.



SACS-AutoPIPE Intraoperability

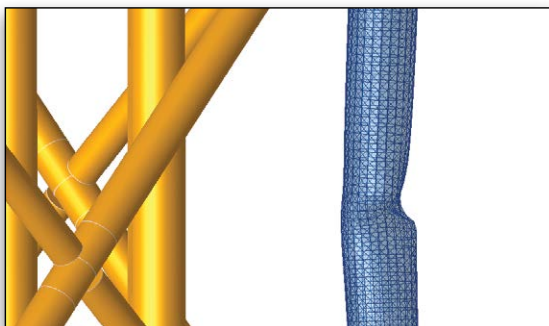
The SACS and AutoPIPE® interface integrates piping design, pipe stress, and structural analysis. It allows users to automatically transfer pipe support loads onto the structural model for analysis, and structural model back to the piping model for visualization, engineering coordination, and combined stiffness model for more accurate pipe support loads. Workflows to add pipe loads to offshore structures removes the requirement for manual data entry. The interface provides realistically engineered designs, and avoids the manual transfer of support loads between the piping and structural models.



SACS – Combined piping and structural model (secondary steel in green)

SACS-MOSES Intraoperability

SACS structural models can be imported into MOSES for offshore installation analysis. Generating SACS TOW files from MOSES allows automatic generation of inertial based acceleration loading and RAOs. Enhancing collaboration across structural and naval architectural teams improves efficiency, reduces rework and project delays. For an improved 3D hydrodynamic response, users can run a MOSES launch analysis directly from SACS Executive using SACS input for a launch analysis.

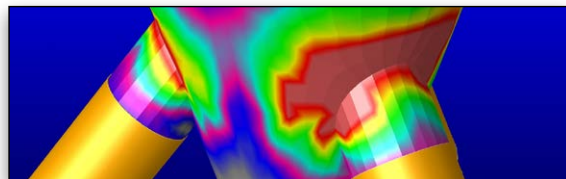


Ship impact analysis.

The following add-on modules can be combined with Offshore Structure Ultimate for comprehensive offshore capabilities.

Pile Structure Design Soil/Pile/Structure Interaction Analysis

This nonlinear add-on package permits nonlinear soil/pile/structure interaction analysis of fixed offshore structures with multiple fixed supports using the PSI program modules. It requires the use of the Offshore Structure, Offshore Structure Advanced, or Offshore Structure Ultimate package.



Parametric study for the fatigue design of a wind turbine transition piece.

Collapse Plastic Nonlinear Add-on

Perform advanced plastic analysis using of the Offshore Structure, Offshore Structure Advanced, or Offshore Structure Ultimate package.

Collapse Advanced Plastic Nonlinear Add-on

Perform advanced nonlinear plastic analysis including pushover, ship impact, dropped object, and blast analysis.

The Collapse Advanced program supports large deflections, large rotations, arc length iterations, automatic sub-incrementations and torsional buckling. It requires the use of Offshore Structure, Offshore Structure Advanced, or Offshore Structure Ultimate package.

Fatigue Advanced – Dynamic Response Fatigue Package with Dynamic Response

This package contains the modules required to determine the wind fatigue damage of a dynamic system. This package contains DYNPAC, Fatigue, Interactive Fatigue, and Dynamic Response. It requires the use of the Offshore Structure, Offshore Structure Advanced, or Offshore Structure Ultimate package.

Fatigue Advanced – Wave Response Fatigue Package with Wave Response

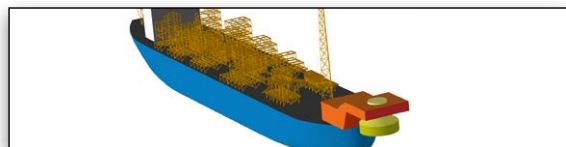
This package contains the modules required to determine the fatigue damage of a dynamic system subject to wave loads. This package contains DYNPAC, Fatigue, Interactive Fatigue, and Wave Response. It requires the use of the Offshore Structure, Offshore Structure Advanced, or Offshore Structure Ultimate package.

Fatigue Ultimate Fatigue Package with Wave Response and Dynamic Response

This package contains the modules required to perform any dynamic deterministic, time history, or spectral fatigue analysis. It requires the use of the Offshore Structure, Offshore Structure Advanced, or Offshore Structure Ultimate package.

Wind OpenWindPower™ Fixed Foundation

The OpenWindPower Fixed Foundation is comprised of the following packages: Offshore Structure Ultimate, Pile Structure Design, Collapse, and Fatigue Ultimate. The package also contains the SACS interfaces to the GH Bladed and FAST wind turbine aero-elastic modules.



Data interchange with MOSES for FPSO topsides analysis.

The following software modules extend the functionality of any of the three offshore structure packages and are either included within those packages or available as add-on modules. Please consult your Bentley representative for product specifics relative to your needs.

SACS Executive

Common Interface to Program Suite

- Controls and connects all elements of the SACS system
- Launches all SACS interactive programs
- Executes all batch program analyses
- Allows access to all SACS system configuration settings, including system file location and security key settings
- Includes command line help and power buttons for the most commonly executed tasks
- Specifies analysis options without changing data input file
- Generates SQLite database for reporting in Excel and other programs
- Integrated with ProjectWise® for sharing of all projects files across multiple sites

Precede

Interactive Full Screen Graphics Modeler

- Model generation capabilities include geometry, material and section properties, and loading
- Automatic input error detection
- Maintains data backup
- Beam and/or finite element modeling including plate and shell elements
- Automatic offshore jacket and deck generation
- User-defined input units
- Cartesian, cylindrical, or spherical mesh generation
- Automatic weight or load generation including gravity, pressure, and skid mounted equipment loads
- Seastate data generation capabilities
- Decommissioning capabilities to automatically slice structure by volume into smaller components of given weight and center of gravity
- Interactive GUI to mesh stiffened joints for automatic stress concentration factor (SCF) extraction and strength analysis

- Extensive plotting and reporting capabilities
- Code check parameter generation including K-factors and compression flange unbraced lengths
- Allows SACS model files to be converted into 3D SAT file format compatible with AutoCAD, and other CAD systems
- Allows the definition of sets comprised of joints, members, plates, shells, and solid elements
- Allows views to be saved as models
- Physical member support capabilities
- Professional and other ACIS-enabled CAD packages
- Supports full 3D geometry and section properties
- Allows SACS model files to be ported directly to a PDMS macro file, which creates the 3D model in PDMS
- Supports PDMS section libraries in addition to creating PDMS sections for sections defined in the SACS model
- Logging functionality
- ISM Export to ProSteel and other steel detailing systems
- Integrated with ProjectWise for sharing all projects files across multiple sites

Data Generator

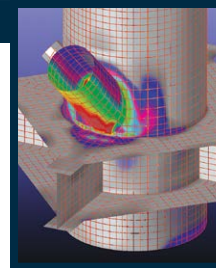
Interactive Data Generation for all Programs

- Intelligent full screen editor that labels and highlights data fields and provides help for data input
- Form-filling data input available as well as full screen mode
- Automatic data checking
- Integrated with ProjectWise for sharing of all projects files across multiple sites

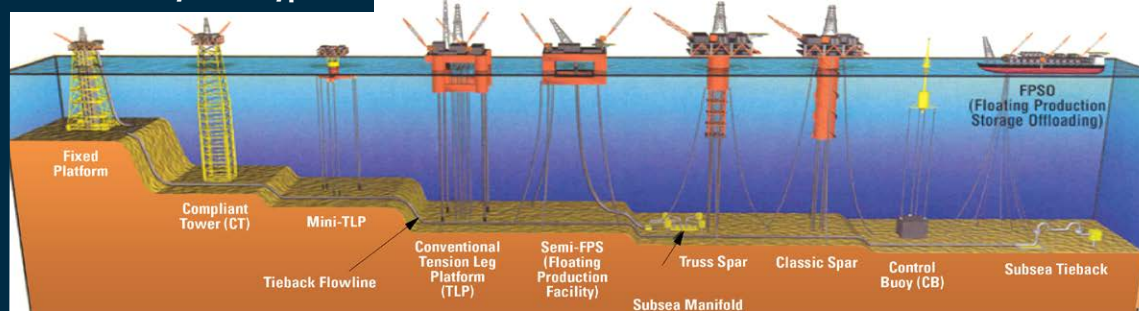
Seastate

Environmental Loads Generator

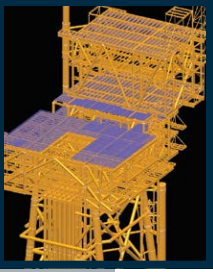
- Ability to view plot files on screen
- Sends viewed plots to printer/plotter
- Supports HP-GL, Postscript, DXF, Windows devices
- Metafile (WMF), and SACS NPF plot file
- Allows plot size, character size, margins, and formats
- Ability to modify chart settings
- Full implementation of API 21st edition



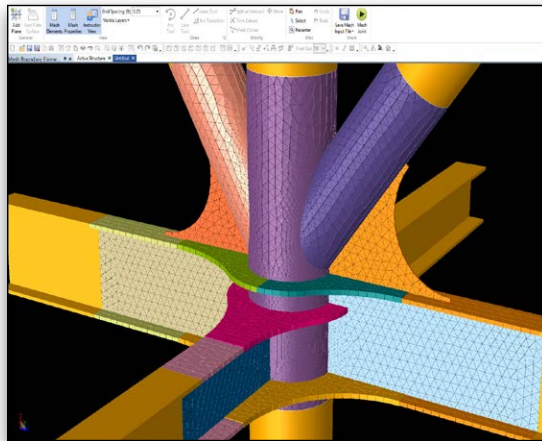
Offshore System Types



SACS has applications for all types of offshore structures and vessels



- Supports five wave theories
- Current included or excluded
- Generates load due to wind, gravity, buoyancy, and mud flow
- Marine growth, flooded, and nonflooded members
- RAO and acceleration loading including nonstructural weights
- Moving loads generation
- Diameter, Reynolds number, and wake encounter effects dependent drag and inertia coefficients
- Weight load cases
- Forces on nonstructural bodies
- Deterministic and random wave modeling for dynamic response
- Member hydrodynamic modeling for static and dynamic analysis
- Corrosion modeling



Graphical user interface for the design and automatic meshing of stiffened nontubular joints.

SACS IV Solver

Static Beam and Finite Element Analysis

- Beam elements including tubulars, tees, wide flanges, channels, angles, cones, plate and box girders, stiffened cylinders, and boxes
- Solid and plate elements (isotropic and stiffened)
- Discrete Kirchoff Theory (DKT) thin-plates
- Isoparametric 6-, 8-, and 9-node shell elements
- Library of AISC, U.K., European, German, Chinese, and Japanese cross sections, as well as user-defined libraries
- Member, plate, and shell local and global offsets
- Beam and finite element thermal loads
- Elastic supports defined in global or reference joint coordinate system
- Specified support joint displacements
- Unlimited load cases

- P-delta effects
- Master/slave DOF

Post

Beam and Finite Element Code Check and Redesign

- Beam and plate element code check and redesign
- API (including 21st edition), AISC, LRFD, Norsok, Eurocode 3, Canadian, DNV, British Standards, and Danish DS449 code check
- Plate panel checks in accordance to DNV-RP-C201
- Creates updated model with redesigned elements
- Modify code check parameters
- Load combination capabilities
- Supports codes from 1977 to present
- Detailed and summary reports
- Hydrostatic collapse analysis
- Span (multimember effects)
- ISO 19902
- Automatic extrication of SCFs from a finite element analysis of a meshed joint

Joint Can

Tubular Joint Code Check and Redesign

- Present and past codes including latest API 21st edition, supplement 2, and LRFD, Norsok, DS449, and Canadian
- API earthquake and simplified fatigue analysis
- ISO 19902
- Connection strength (50 percent) check
- Overlapping joints analyzed
- Minimum and extreme seismic analysis

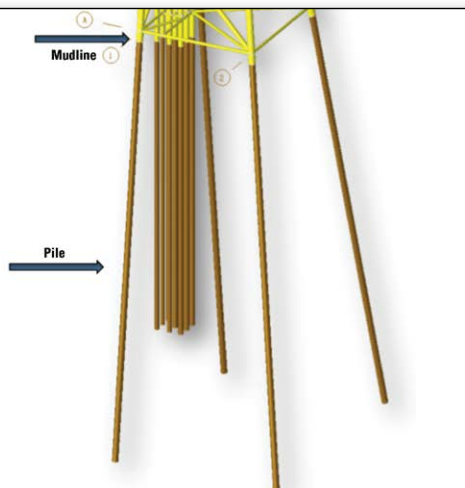
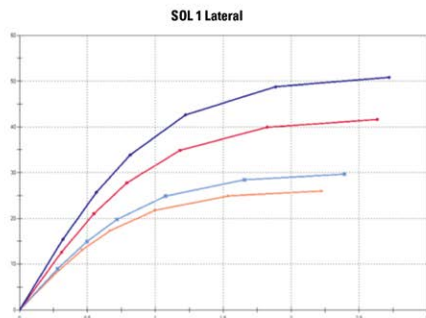
Postvue

Interactive Graphics Post Processor

- Interactive member and tubular joint code check and redesign, with the option to print code check details for latest AISC, ASD and LRFD, API, ISO 19902 codes
- Display shear and bending moment diagrams
- Display rendered deflected shapes for static and dynamic analyses
- Animate deflected and modal shapes
- Color plate stress contour plots
- Code check and redesign by individual or group of elements
- Supports same codes as post module

PSI:

- *API P-Y / T-Z Soil*
- *API Adhesion Soil*
- *User Defined P-Y / T-Z Soil*
- *User Defined Adhesion Soil*



- Extensive reporting and plotting capabilities
- Color-coded results and unity check plots
- Creates updated input model file for re-analysis
- Labels UC ratio, stresses, and internal forces on elements

Concrete**Reinforced Concrete Code Check and Redesign**

- Rectangular, Circular, Tee, and L cross sections
- Beam, bi-axial beam-column, slab, and wall elements supported
- Multiple reinforcement patterns can be specified
- Code check per ACI 318-89 (revised 1992)
- Shear reinforcement check and redesign
- Reinforcement development length check
- Deflection and creep calculation
- Second order/P-delta analysis capabilities

Fatigue**Fatigue Life Evaluation and Redesign**

- Spectral, time history, and deterministic fatigue analysis
- Cyclic stress range calculation procedures include wave search, curve fit, and interpolation
- SCF calculations recommended by API (including 21st edition supplements), HSE, DNV, DS449, and Norsok Codes
- Automatic redesign
- API (including 21st edition supplements), AWS, HSE, and Norsok thickness dependent recommended S-N curves
- Multiple run damage accumulation
- Pierson-Moskowitz, JONSWAP, Ochi-Hubble double peak, simplified double peak, and user-defined spectra
- Automated or user-specified connection details

- Pile fatigue analysis
- Creates wave spectra from scatter diagram
- Uses Paris equation to predict crack growth rate due to cyclic stresses
- Load path dependent joint classifications
- Includes wave spreading effects
- Reservoir (rain flow) cycle counting method
- ISO 19902

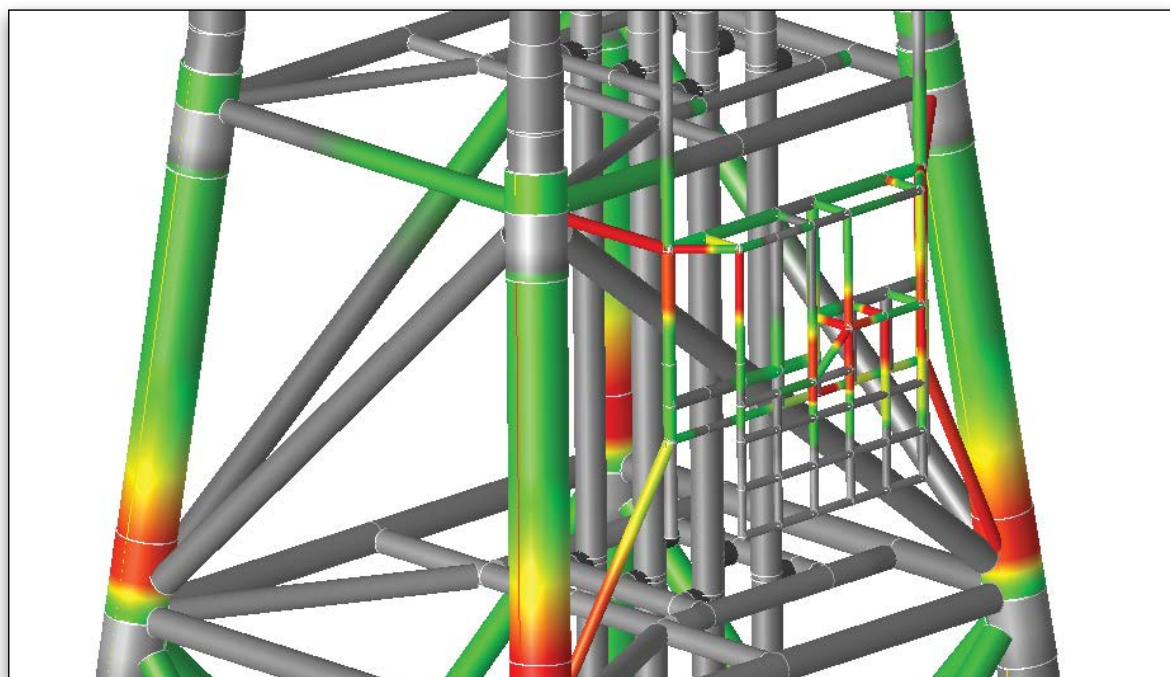
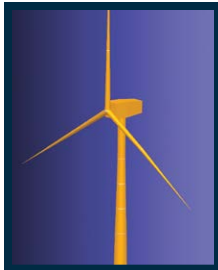
Interactive Fatigue**Interactive Fatigue Life Evaluation**

- Shows the 3D view of the connection and allows braces to be selected with the mouse
- Reads connection defaults when joint and/or brace is/are selected, thus eliminating the need to calculate and display SCFs before viewing capacity or modifying properties
- Recognizes all SCF and S-N options available in the batch program
- Allows SCF theory to be changed for any type connection, including in-line connections and connections with user defined SCFs
- Reports have been expanded and reworked to make them easier to read
- Reports and plots can be displayed on the screen and/or saved to a file
- Automatic redesign

GAP**Nonlinear Analysis with One-way Elements**

- Accurate simulation of load out or transportation analysis using one-way elements
- Tension or compression gap elements with initial gap
- General nonlinear elements
- Friction element





Nonlinear elasto-plastic deformations.

PSI

Nonlinear Soil, Pile, and Structure Interaction

- Beam column effects included
- Nonuniform piles
- P-Y and T-Z curves, axial adhesion and springs
- API P-Y, T-Z, skin friction and adhesion data generated from soil properties per API
- Full structural analysis and pile code check API, LRFD, Norsok, HSE, DS449, Canadian, and DNV
- Offset P-Y & T-Z curves for mudslides
- Full plotting and graphical representation of soil data and results, including stresses, P-Y, T-Z curves
- Soil liquefaction effects

Pile

Isolated 3D Pile Analysis

- Beam column and pile batter effects included
- Uses PSI soil data
- Optional pile head springs
- Specify force at or below pile head
- Specify pile head displacements
- Specified pile head forces or displacements
- Automatic generation of linear equivalent pile stubs for dynamic or static analysis
- Soil liquefaction effects
- Same plotting and code check features as PSI

Superelement

Automated Substructure Creation and Application

- Unlimited number of superelements
- Up to 1,000 interface joints per superelement
- Translation and rotation of superelements
- User-defined stiffness matrices
- Full stress recovery
- Superelements can contain other superelements
- Translation and rotation of superelements

Dynamic Superelement

- Can be based upon either Guyan or Craig-Bampton methods
- Uses Modal Assurance Criterion to check accuracy
- User-defined output format

Combine

Common Solution File Utility

- Combines dynamic and static results from one or multiple solution files
- Combines results from analyses having different member, plates, etc.
- Superimposes mode shapes
- "Worst-case" combination of dead loads with earthquake response
- Determine extreme wave loads from input spectra

Large Deflection (LDF)

Large Deflection Analysis

- Iterative solution for geometric
- Solves plate membrane problems
- Accounts for P-delta effects nonlinearities

Collapse

Nonlinear Collapse Analysis

- Linear and nonlinear material behavior
- Nonlinear springs
- Sequential load stacking capability
- Activate and deactivate elements
- Joint flexibility options
- Impact analysis with automatic unloading, built-in DNV ship indentation curves and energy absorption functionality
- Load cases may contain loading and/or specified displacements
- Includes geometric nonlinearities
- Plastic members and finite elements
- Includes piles with nonlinear soils and plasticity
- Soil liquefaction effects
- Plastic DKT plates

Dynpac

Dynamic Characteristics

- Householder-Givens solution
- Guyan reduction of non-essential degrees of freedom
- Lumped or consistent structural mass generation
- Automatic virtual mass generation
- Complete seastate hydrodynamic modeling

- User input distributed and concentrated mass
- Nonstructural weight modeling
- Full 6 DOF modes available for forced response analysis

Wave Response

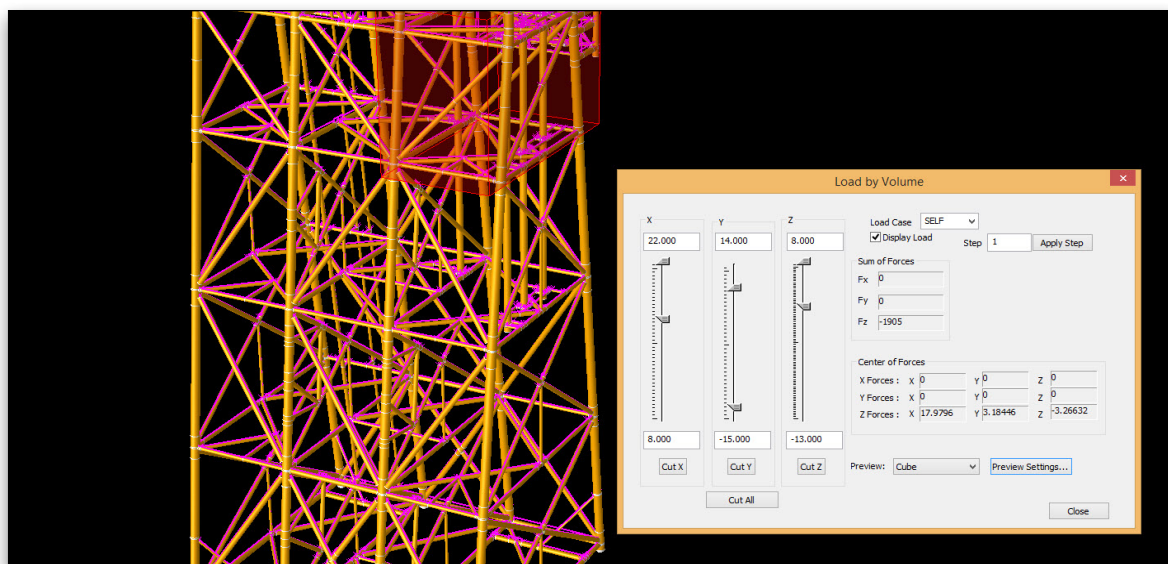
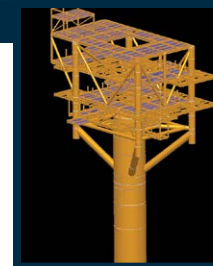
Dynamic Wave Response

- Deterministic and random waves
- Pierson-Moskowitz, Jonswap, Ochi-Hubble, and user wave spectra
- Harris, Von Karmon, and Kaimal wind spectra
- Fluid-structure relative velocity and acceleration accounted for "Modal Acceleration" and nonlinear fluid damping
- Closed form steady state response in the frequency domain
- Equivalent static load output for accurate stress recovery
- Zero crossing and RMS responses
- Time history analysis of wave and wind and time history load
- Buoyancy dynamic loads included
- Stress, internal load, base shear, and overturning moment transfer function plots available
- Full coupling with Fatigue program
- Elastic dynamic response of floating structures including stingers
- Input and output Power Spectral Densities with Probability Distributions
- Special features for wind turbine analysis

Dynamic Response

General Dynamic Response and Earthquake Analysis

- Frequency domain analysis
- Time history, response spectrum or PSD-base driven input for seismic analysis
- Force time history and harmonic-force driven analysis
- SRSS, CQC and peak modal combinations



Precede graphics modeler decommissioning capabilities.

System Requirements

Processor:

CPU: Pentium 4 or higher

Operating System:

Windows 7, 8/8.1, 10

RAM:

Minimum 512 MB SACS performance is dependent on model size and resources available

Hard Disk:

Minimum 800 MB partition for SACS installation

Display:

Graphics card supporting Open GL
128 MB RAM or greater video card with 1280x1024 or higher video resolution

Network:

A network connection is required.
100 Base-T or greater Local Area Ethernet NetworkTCP/IP network protocol is supported

Find out about Bentley at: www.Bentley.com/Offshore

Contact Bentley

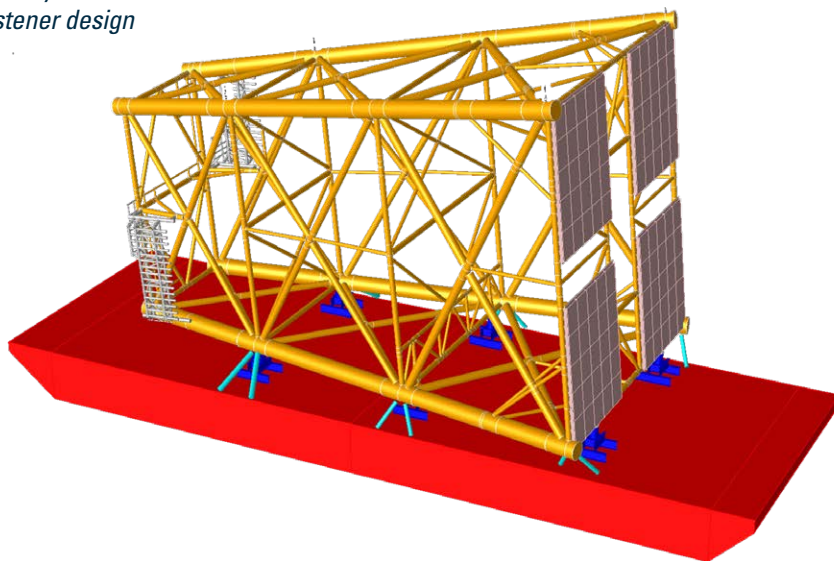
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Global Office Listings

www.bentley.com/contact

Tow Structural Analysis

- Combine multiple common solution files
- Static analysis with nonlinear GAP elements
- Seafastener design



- API response spectra library and user input spectra
- Wind spectral loading capability
- Structural and fluid damping
- Vibration analysis with multiple input points with user specified frequencies and phasing
- General periodic forces decomposed by Fourier analysis
- Ice dynamics analysis
- Engine/compressor vibration analysis
- Response spectrum output at any joint
- Equivalent static load and incremental load output resulting from earthquake, ship impact, dropped object and blast analysis. This loading can be used for subsequent linear static analysis or for nonlinear collapse analysis
- Ship impact analysis
- Dropped object analysis

MTO

Material Take off, Weight Control, and Cost Estimation

- Member lengths including cuts
- Steel tonnage and CG location
- Material list, cost estimate, and weight control reports
- Weld volume requirements and cost
- Required protective anodes and cost
- Surface area calculations by elevation
- Anode calculation in accordance to NACE SP0176-2007 (formerly RP0176-2003) and DNV-RP-B401

Tow

Transportation Inertia Load Generator

- Input motion for six degrees of freedom
- Output location for selected points
- Automatic weight calculation
- User input member and joint weights
- Generates distributed member and plate loads
- Converts user defined loads into inertias